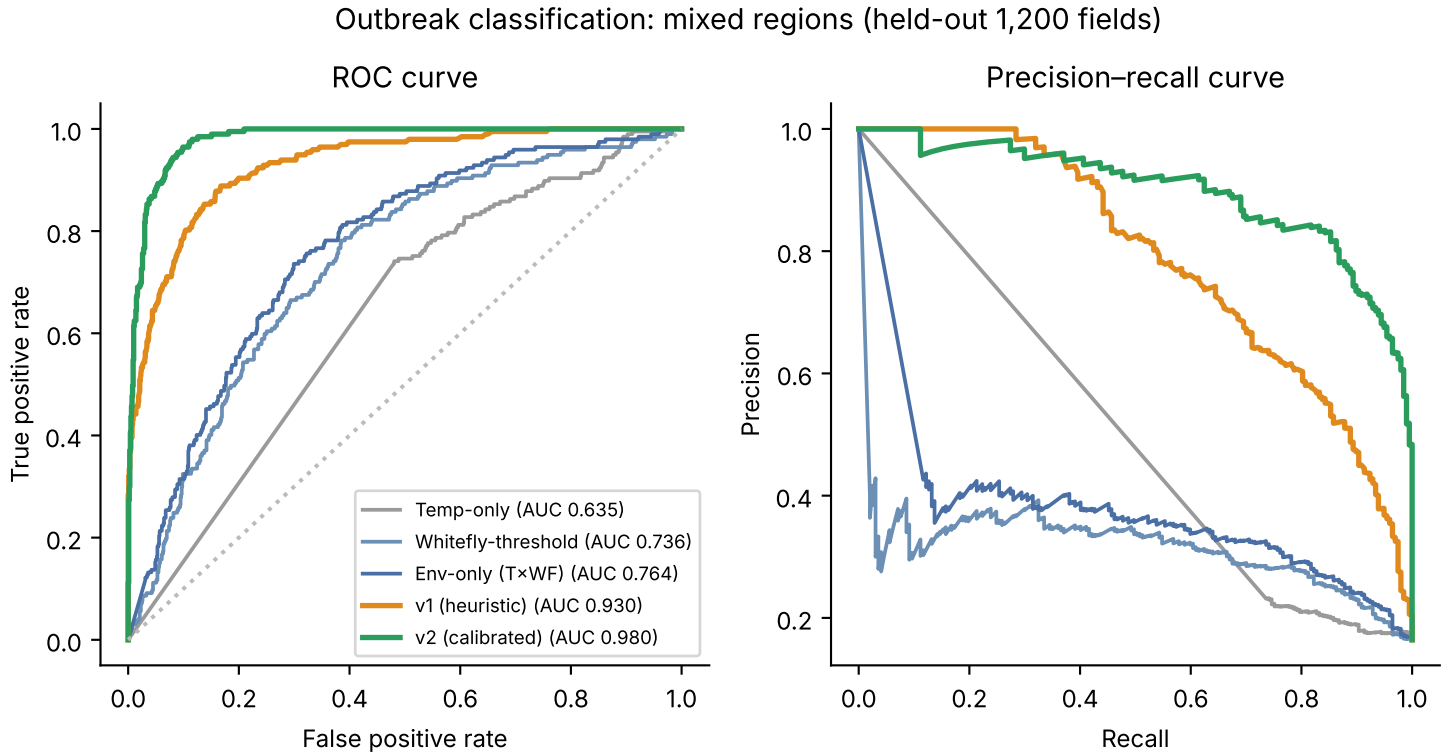
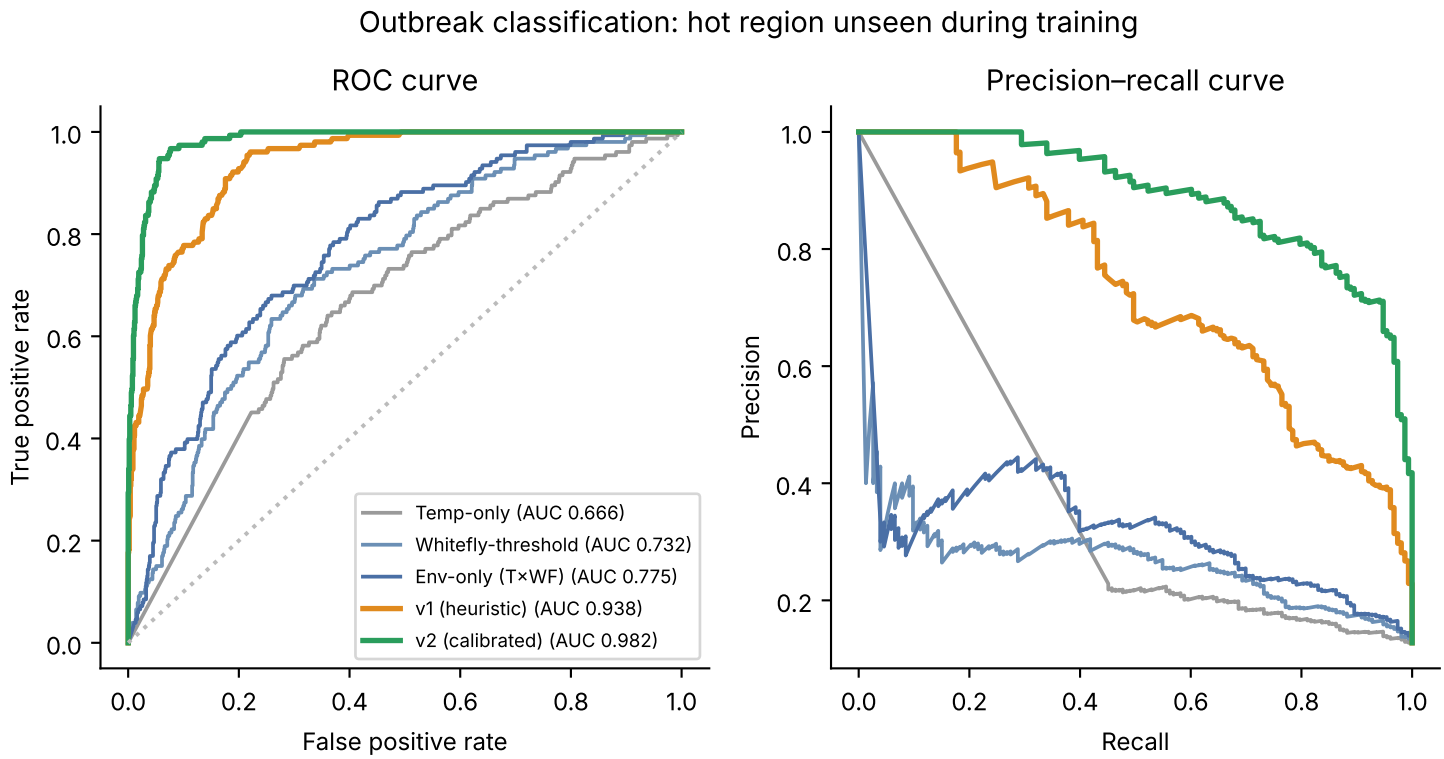
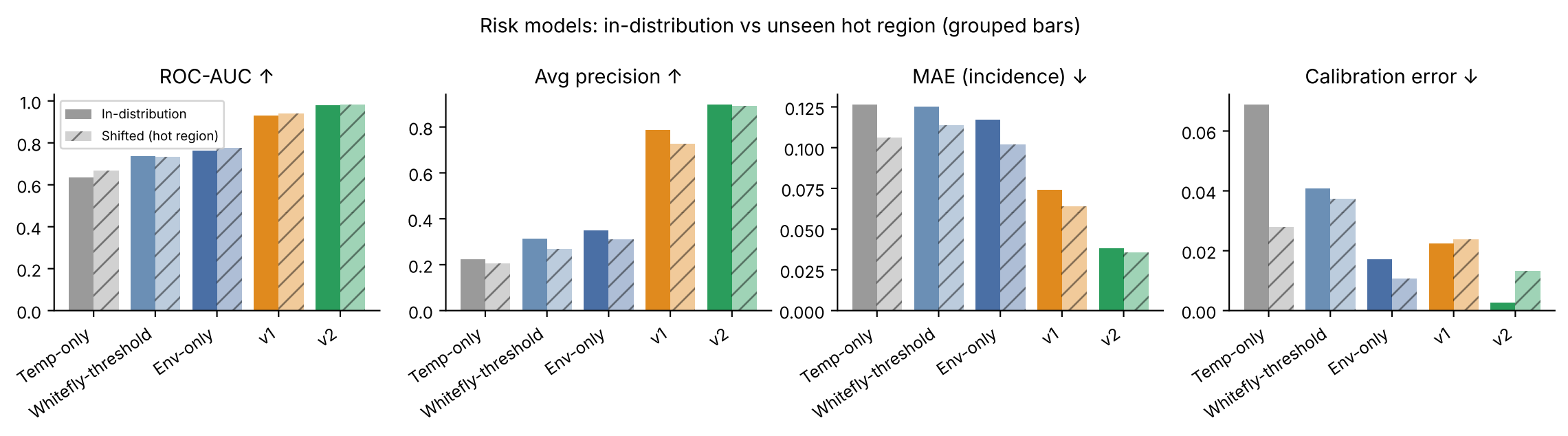
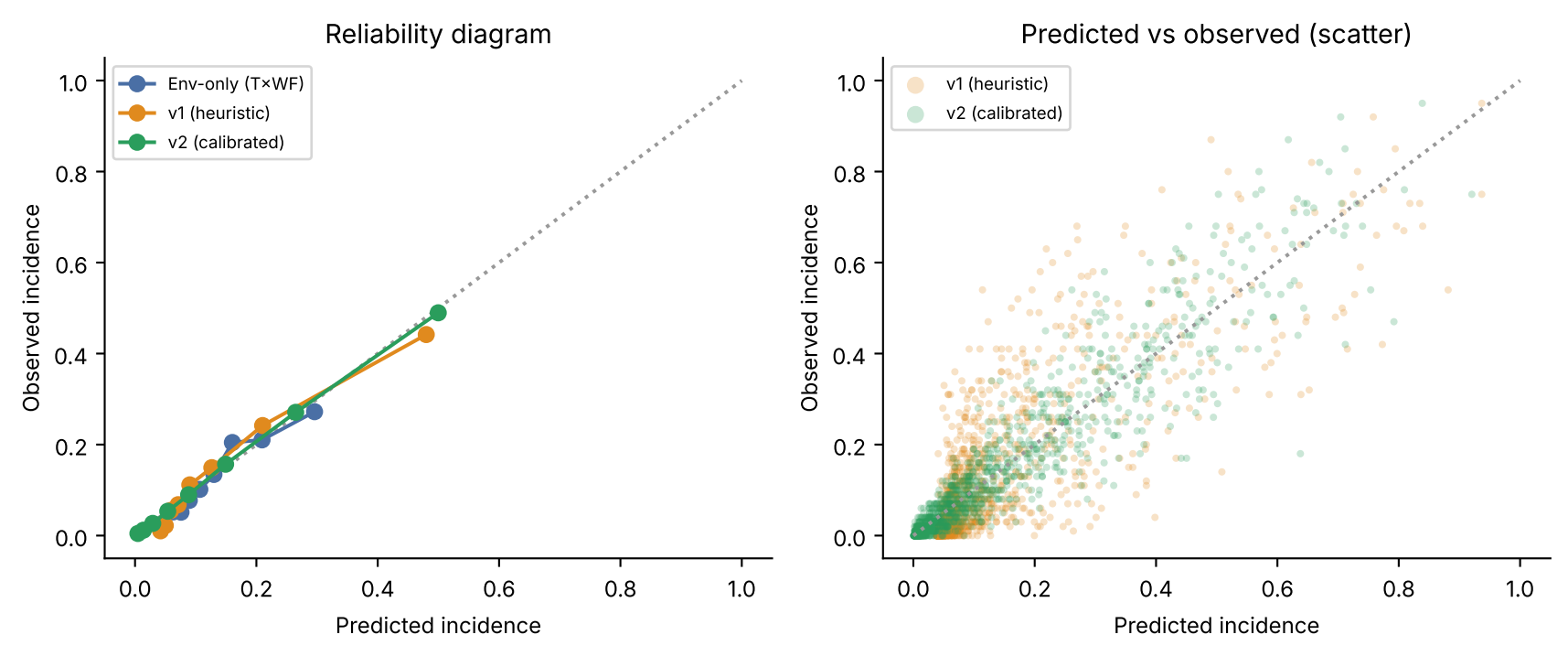
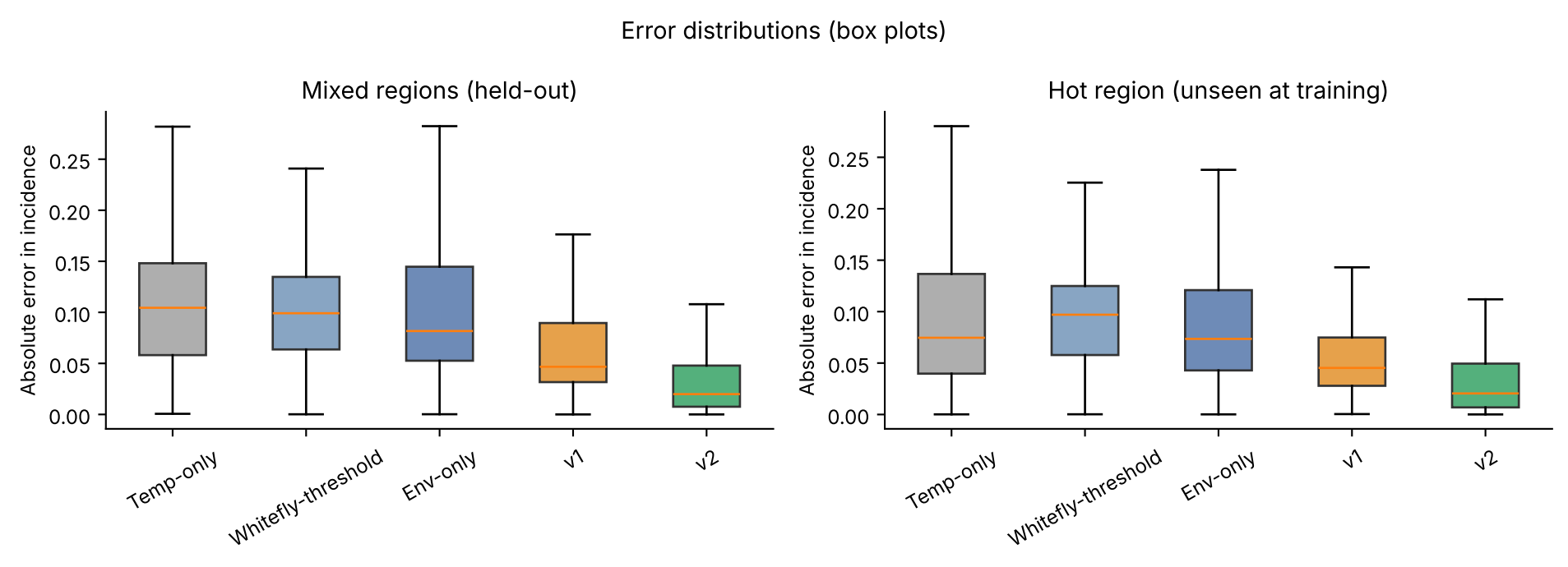
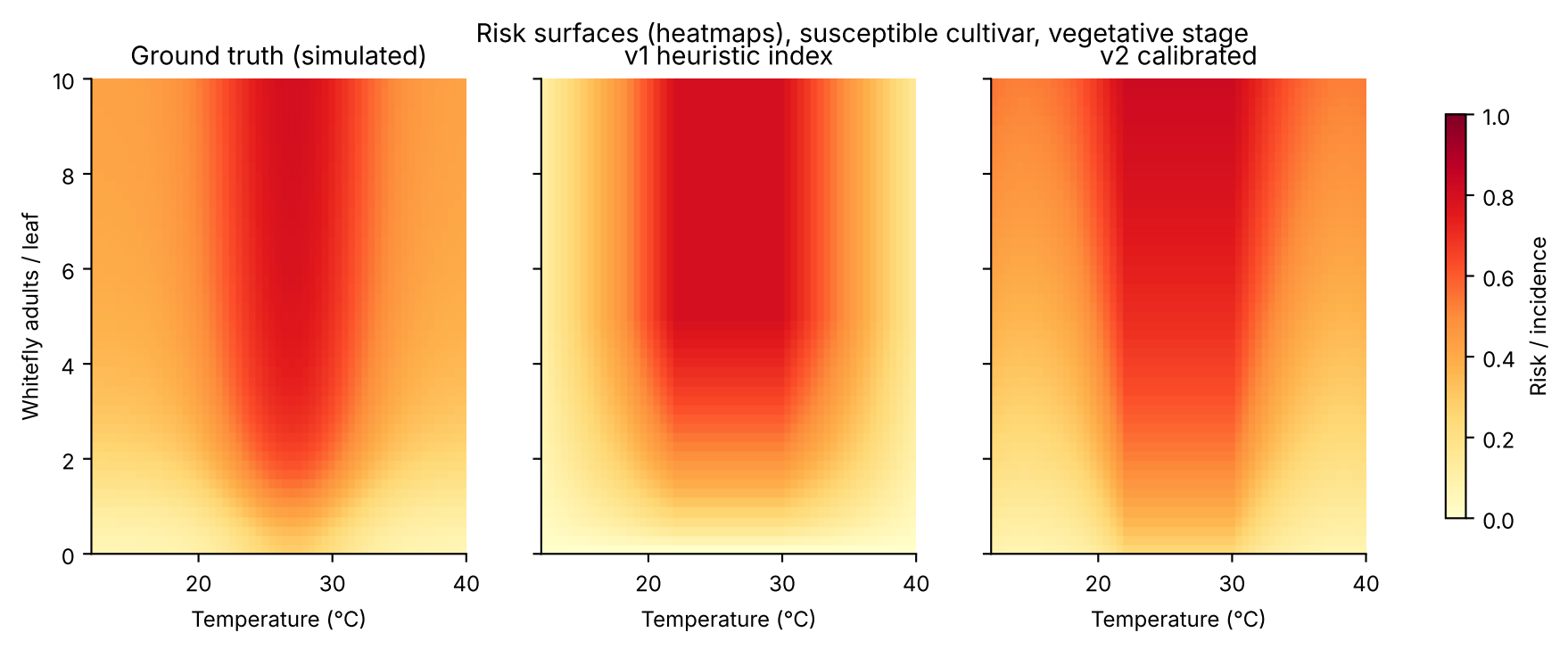

Outbreak classification: mixed regions (held-out 1,200 fields)
ROC curve
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Temp-only (AUC 0.635)
Whitefly-threshold (AUC 0.736)
Env-only (T×WF) (AUC 0.764)
v1 (heuristic) (AUC 0.930)
v2 (calibrated) (AUC 0.980)
Precision–recall curve
Precision
Recall

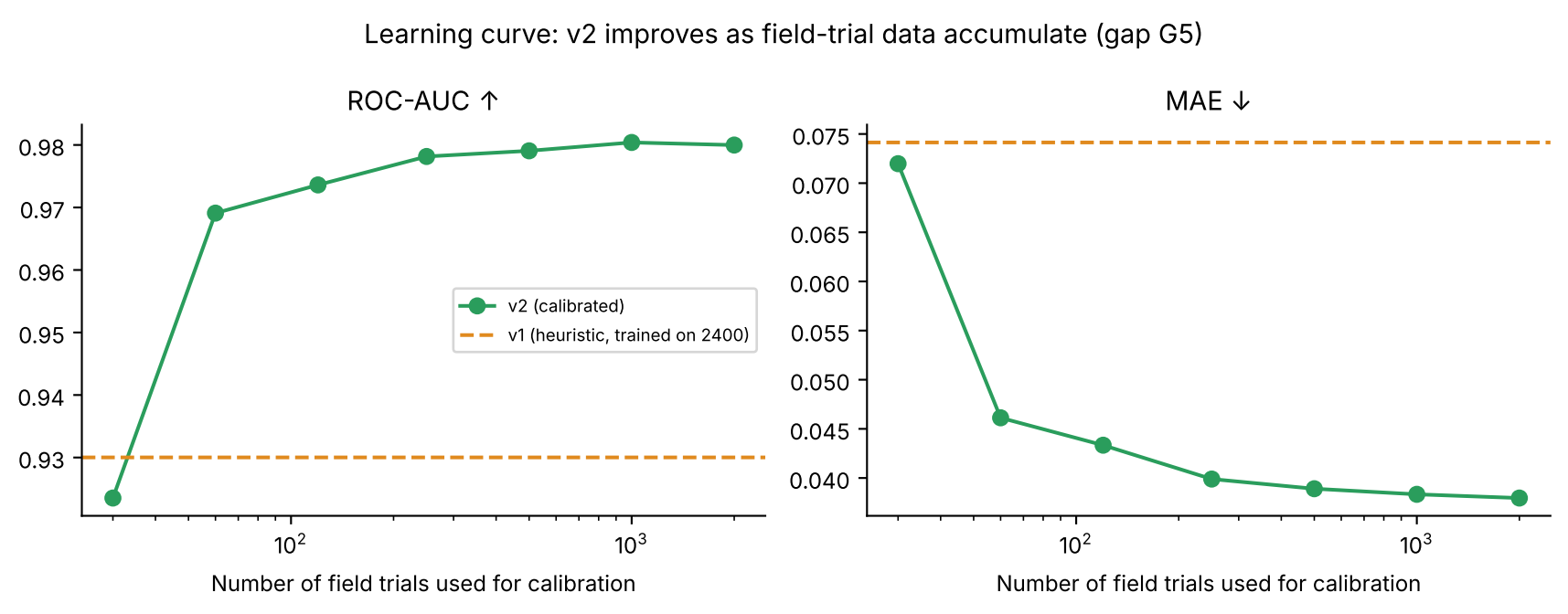

Outbreak classification: hot region unseen during training
ROC curve
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Temp-only (AUC 0.666)
Whitefly-threshold (AUC 0.732)
Env-only (T×WF) (AUC 0.775)
v1 (heuristic) (AUC 0.938)
v2 (calibrated) (AUC 0.982)
Precision–recall curve
Precision
Recall

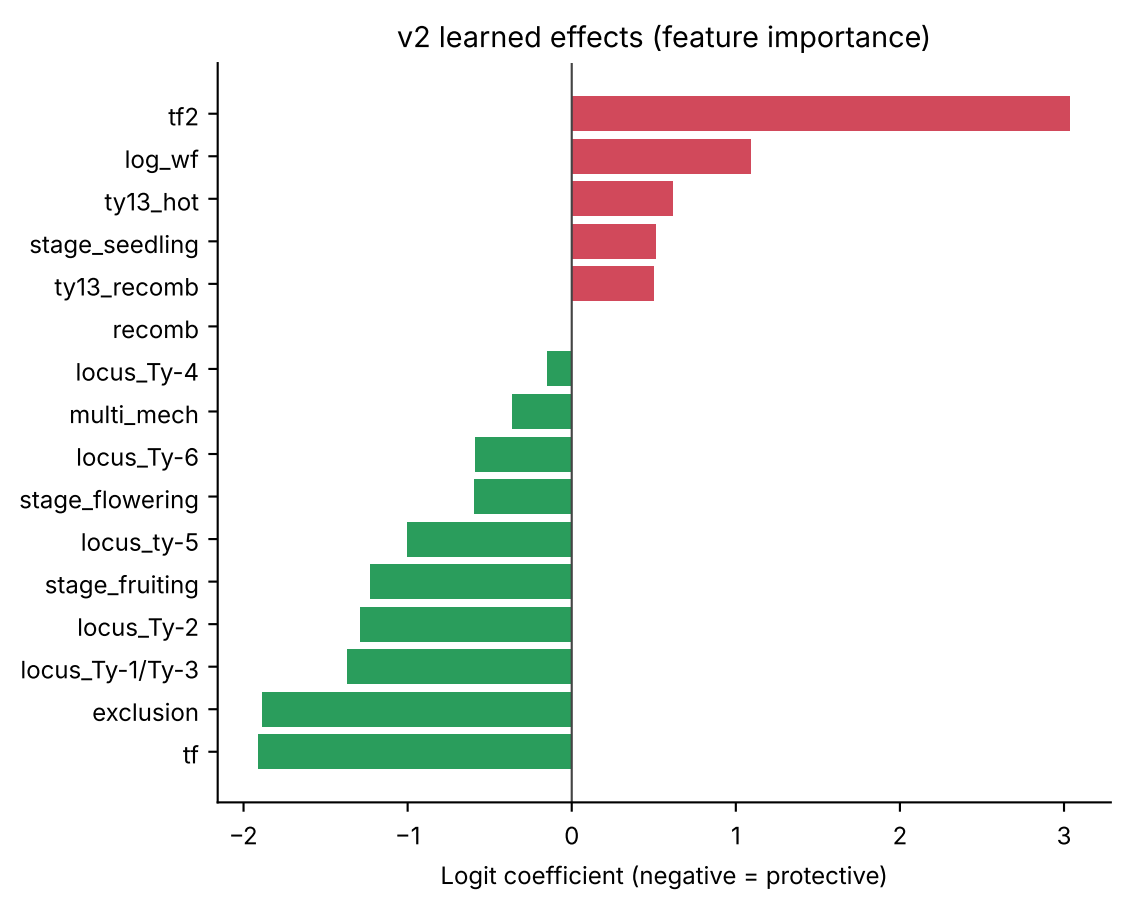

Risk models: in-distribution vs unseen hot region (grouped bars)
ROC-AUC ↑
Avg precision ↑
MAE (incidence) ↓
Calibration error ↓
In-distribution
Shifted (hot region)
Temp-only
Whitefly-threshold
Env-only
v1
v2

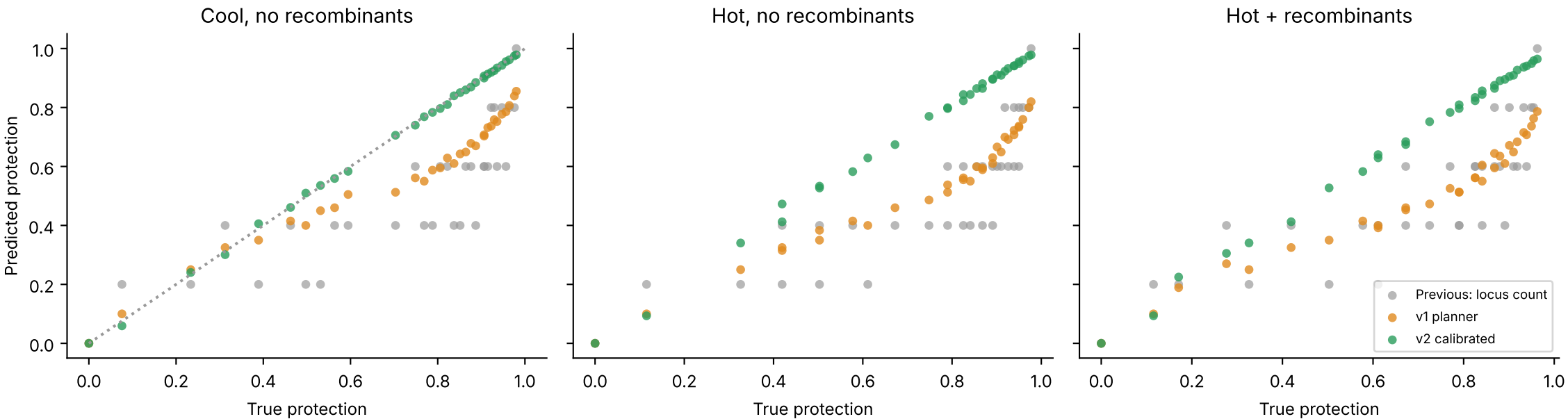

Reliability diagram
Env-only (T×WF)
v1 (heuristic)
v2 (calibrated)
Observed incidence
Predicted incidence
0.0
0.2
0.4
0.6
0.8
1.0
Predicted vs observed (scatter)
v1 (heuristic)
v2 (calibrated)
Observed incidence
Predicted incidence

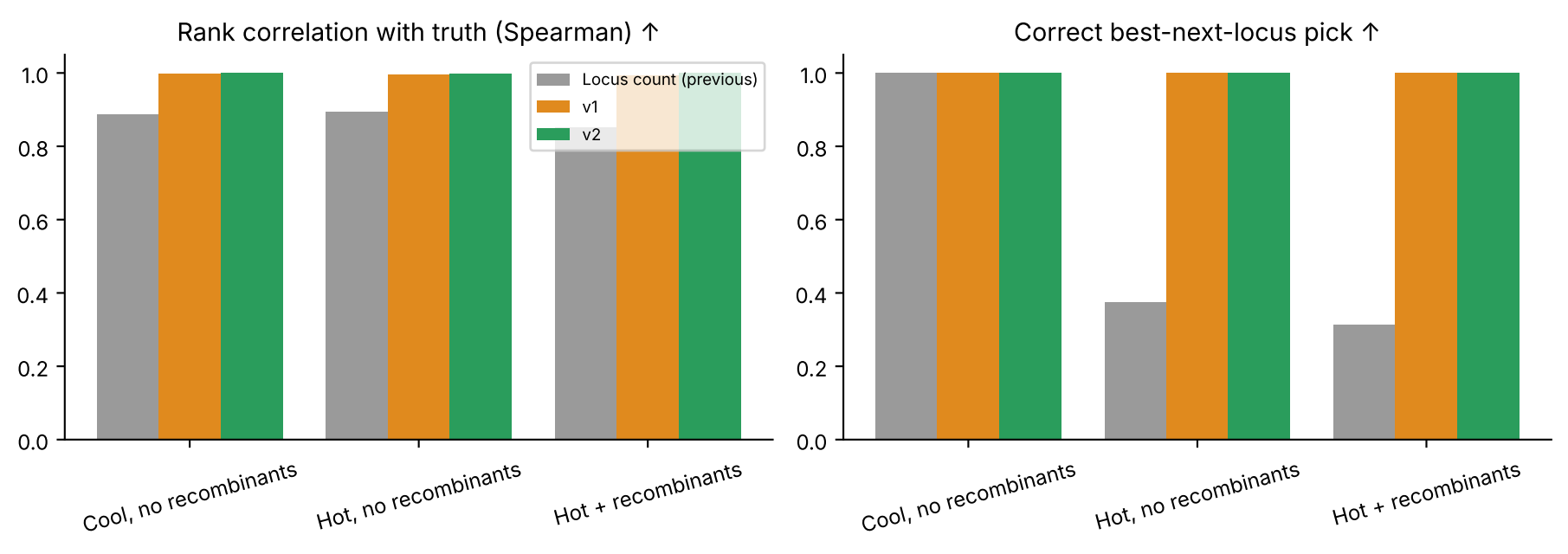

Error distributions (box plots)
Mixed regions (held-out)
Hot region (unseen at training)
Absolute error in incidence
0.00
0.05
0.10
0.15
0.20
0.25
Temp-only
Whitefly-threshold
Env-only
v1
v2

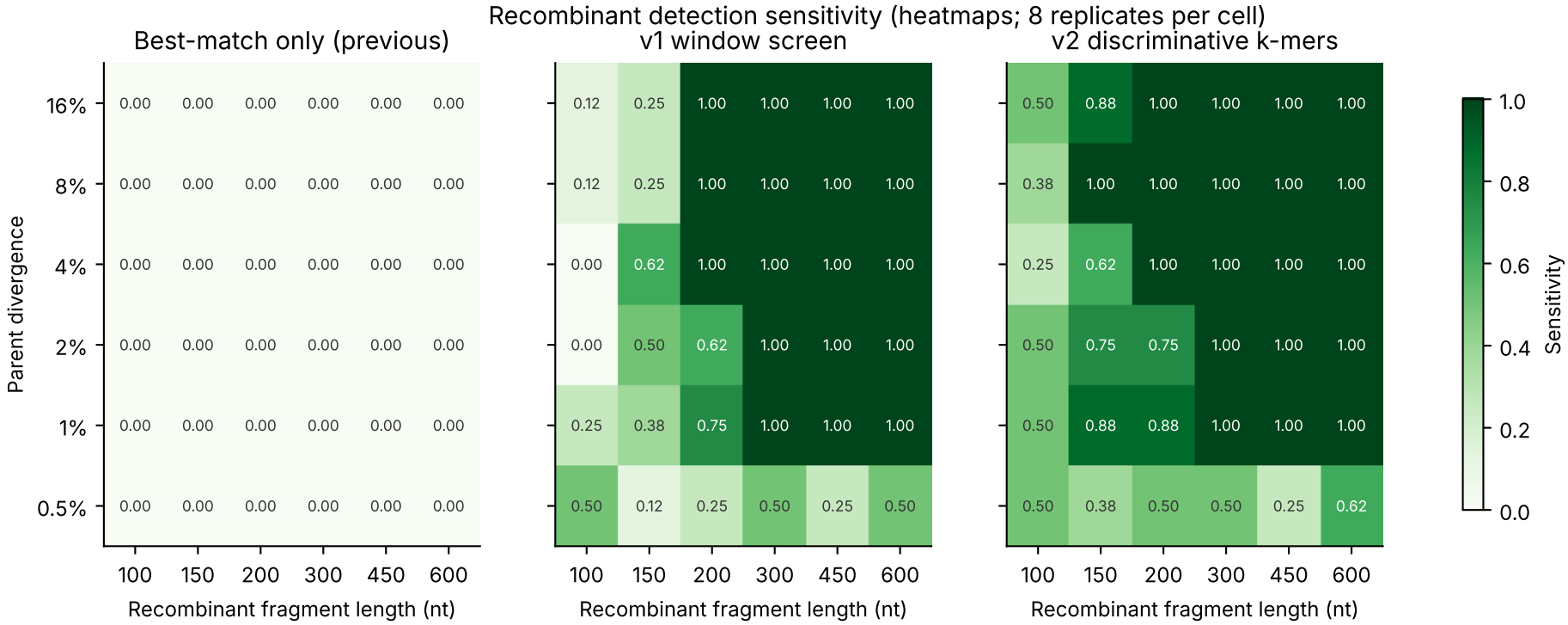

Risk surfaces (heatmaps), susceptible cultivar, vegetative stage
Ground truth (simulated)
v1 heuristic index
v2 calibrated
Whitefly adults / leaf
0
2
4
6
8
10
20
30
40
Temperature (°C)
Risk / incidence
0.0
0.2
0.4
0.6
0.8
1.0

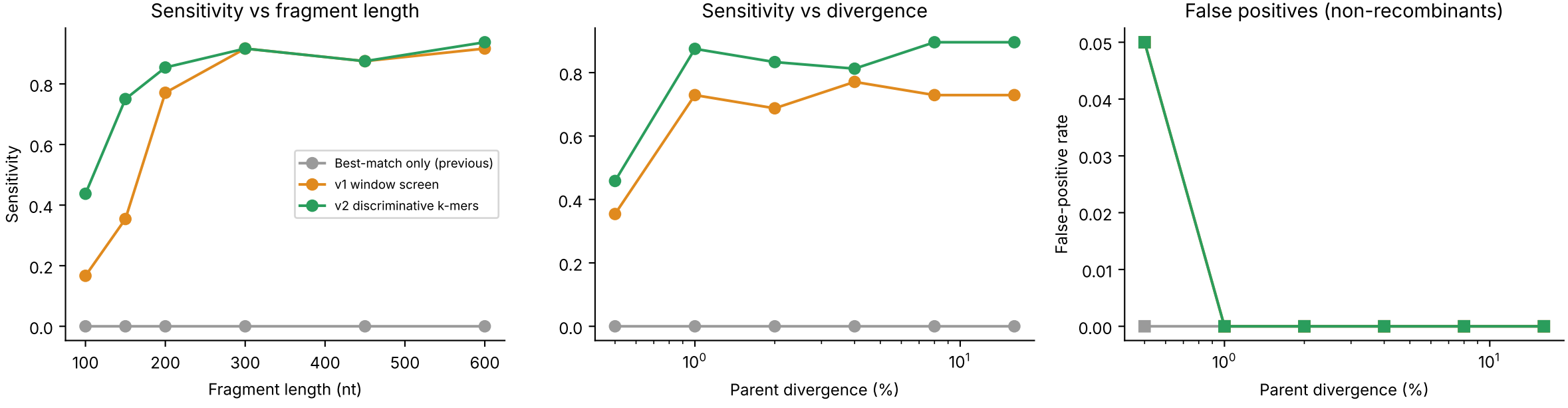

Learning curve: v2 improves as field-trial data accumulate (gap G5)
ROC-AUC ↑
MAE ↓
0.98
0.97
0.96
0.95
0.94
0.93
0.075
0.070
0.065
0.060
0.055
0.050
0.045
0.040
v2 (calibrated)
v1 (heuristic, trained on 2400)
10^2
10^3
Number of field trials used for calibration
Number of field trials used for calibration

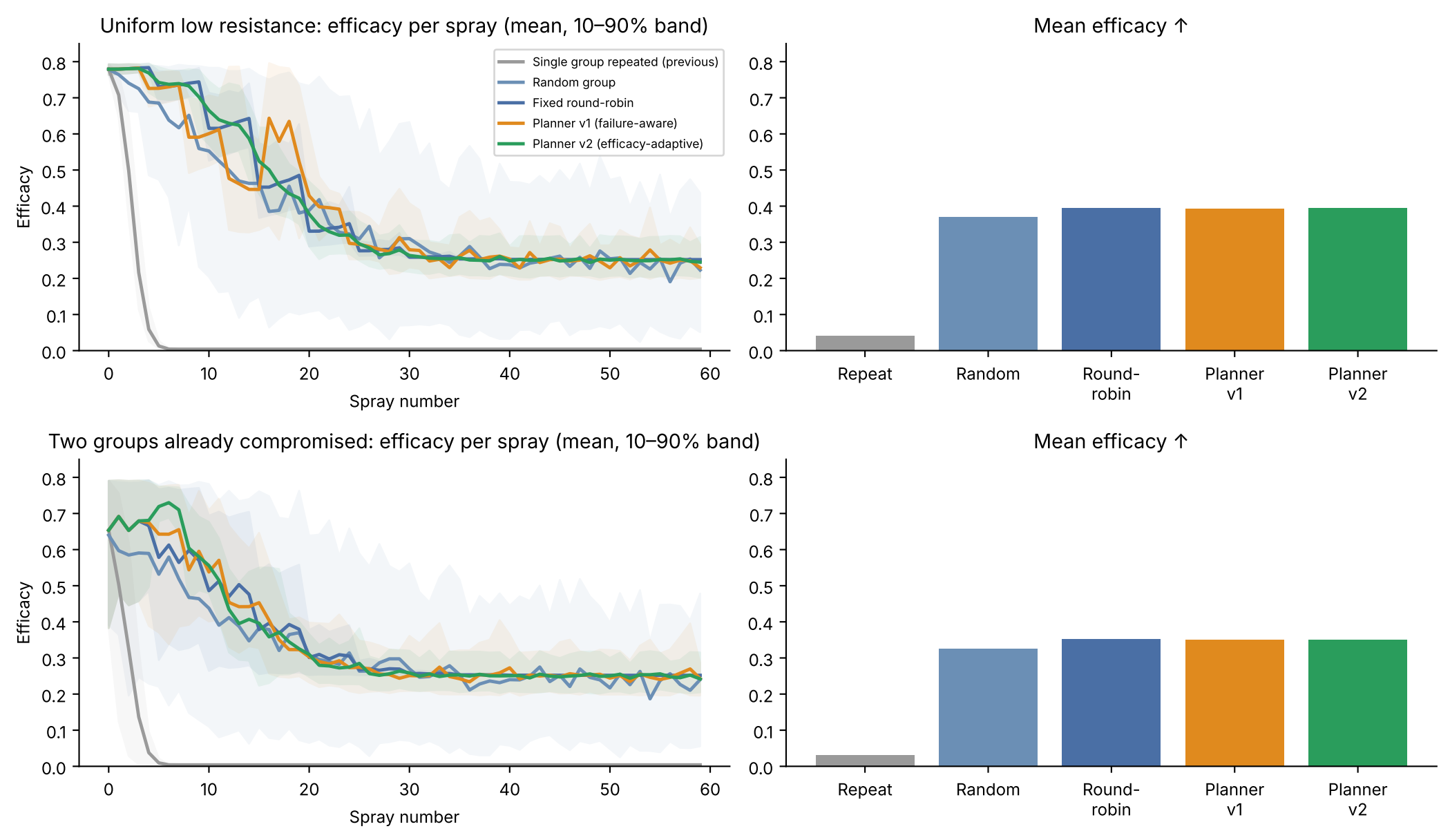

v2 learned effects (feature importance)
tf2
log_wf
ty13_hot
stage_seedling
ty13_recomb
recomb
locus_Ty-4
multi_mech
locus_Ty-6
stage_flowering
locus_ty-5
stage_fruiting
locus_Ty-2
locus_Ty-1/Ty-3
exclusion
tf
−2
−1
0
1
2
3
Logit coefficient (negative = protective)

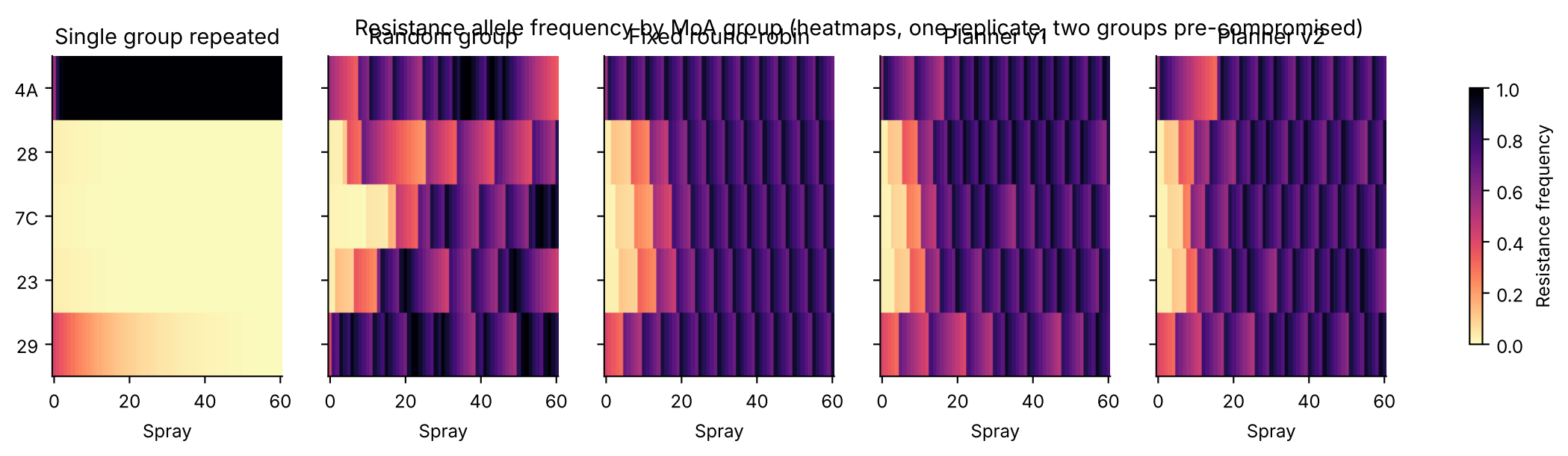

Stacking planner vs truth, all 32 locus combinations (scatter)
Cool, no recombinants
Hot, no recombinants
Hot + recombinants
Predicted protection
True protection
0.0
0.2
0.4
0.6
0.8
1.0
Previous: locus count
v1 planner
v2 calibrated

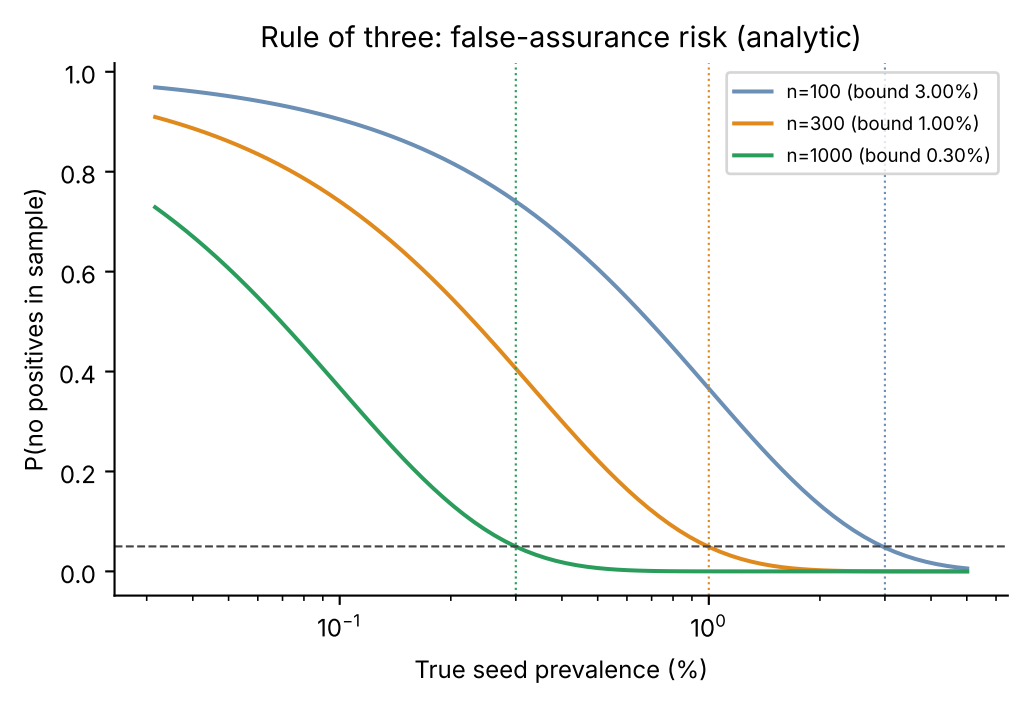

Rank correlation with truth (Spearman) ↑
Correct best-next-locus pick ↑
Locus count (previous)
v1
v2
1.0
0.8
0.6
0.4
0.2
0.0
Cool, no recombinants
Hot, no recombinants
Hot + recombinants

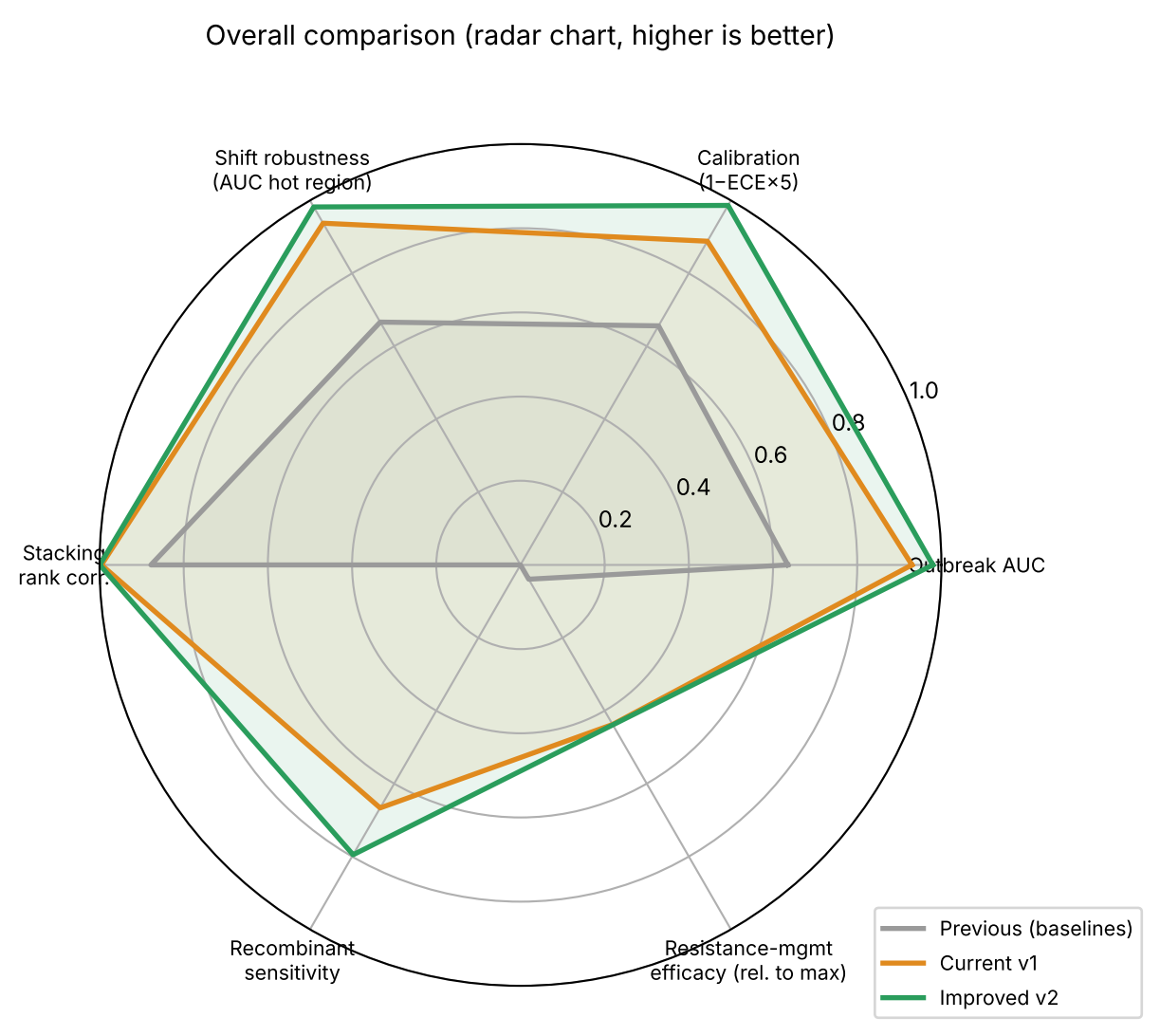

Recombinant detection sensitivity (heatmaps; 8 replicates per cell)
Best-match only (previous)
v1 window screen
v2 discriminative k-mers
Parent divergence
16%
8%
4%
2%
1%
0.5%
0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00
0.12 0.25 1.00 1.00 1.00 1.00
0.12 0.25 1.00 1.00 1.00 1.00
0.00 0.62 1.00 1.00 1.00 1.00
0.00 0.50 0.62 1.00 1.00 1.00
0.25 0.38 0.75 1.00 1.00 1.00
0.50 0.12 0.25 0.50 0.25 0.50
0.50 0.88 1.00 1.00 1.00 1.00
0.38 1.00 1.00 1.00 1.00 1.00
0.25 0.62 1.00 1.00 1.00 1.00
0.50 0.75 0.75 1.00 1.00 1.00
0.50 0.88 0.88 1.00 1.00 1.00
0.50 0.38 0.50 0.50 0.25 0.62
100 150 200 300 450 600
Recombinant fragment length (nt)
Sensitivity
1.0
0.8
0.6
0.4
0.2
0.0


Sensitivity vs fragment length
Sensitivity
Fragment length (nt)
Best-match only (previous)
v1 window screen
v2 discriminative k-mers
Sensitivity vs divergence
Parent divergence (%)
False positives (non-recombinants)
False-positive rate
Parent divergence (%)


Uniform low resistance: efficacy per spray (mean, 10–90% band)
Single group repeated (previous)
Random group
Fixed round-robin
Planner v1 (failure-aware)
Planner v2 (efficacy-adaptive)
Efficacy
Spray number
Mean efficacy ↑
Repeat
Random
Round-robin
Planner v1
Planner v2
Two groups already compromised: efficacy per spray (mean, 10–90% band)
Efficacy
Spray number
Mean efficacy ↑
Repeat
Random
Round-robin
Planner v1
Planner v2


Resistance allele frequency by MoA group (heatmaps, one replicate, two groups pre-compromised)
Single group repeated
Random group
Fixed round-robin
Planner v1
Planner v2
4A
28
7C
23
29
0
20
40
60
Spray
1.0
0.8
0.6
0.4
0.2
0.0
Resistance frequency


Rule of three: false-assurance risk (analytic)
P(no positives in sample)
1.0
0.8
0.6
0.4
0.2
0.0
10^{-1}
10^{0}
True seed prevalence (%)
n=100 (bound 3.00%)
n=300 (bound 1.00%)
n=1000 (bound 0.30%)


Overall comparison (radar chart, higher is better)
Shift robustness
(AUC hot region)
Calibration
(1−ECE×5)
1.0
0.8
0.6
0.4
0.2
Outbreak AUC
Recombinant
sensitivity
Resistance-mgmt
efficacy (rel. to max)
Previous (baselines)
Current v1
Improved v2